Task 10 – Theory versus experiment

The expected pattern is linear growth in the number of comparisons for linear search, logarithmic growth for recursive binary search, and approximately constant average lookup work for a hash table. Real elapsed time also reflects interpreter overhead, memory access, CPU load, data layout and timing noise. Hash-table lookup can be very fast, but building the index takes time and memory. A fair production comparison should measure index construction/update cost and memory as well as query latency. The included graph is a measured sample, not a universal guarantee.